Assignment 4

STAT-540

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1 Exercise 5.1

Show that the scalar SDE

$$dX_t = \alpha X_t dt + \sigma dW_t,$$
$$X_0 = x_0,$$

has the solution

$$X_t = e^{\alpha t} \cdot X_0 + \sigma \int_0^t e^{\alpha(t-s)} dW_s \tag{1}$$

by differentiating X as defined by eqn (1) and showing that X so defined actually satisfies the SDE. Hint: Write eqn (1) as

$$X_t = Y_t + Z_t \cdot R_t,$$

where

$$Y_t = e^{\alpha t} \cdot x_0,$$
$$Z_t = e^{\alpha t} \cdot \sigma,$$
$$R_t = \int_0^t e^{-\alpha s} dW_s,$$

and first compute the differentials dZ, dY and dR. Then use the multidimensional Itô formula on the function $f(y, z, r) = y + z \cdot r$.

Solution:

To solve this first we need to derive the multidimensional Ito's formula as stated in the problem:

$$df(y,z,r) = \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial r} dr + \frac{1}{2} \sum_{i,j} \frac{\partial^2 f}{\partial x_i \partial x_j} dx_i dx_j$$

Then we need to solve for the differentials:

$$dY_t = \frac{d}{dt}(e^{\alpha t} x_0) = \alpha e^{\alpha t} x_0 d_t = \alpha Y_t d_t$$

$$dZ_t = \frac{d}{dt}(e^{\alpha t}\sigma) = \alpha e^{\alpha t}\sigma d_t = \alpha Z_t d_t$$

$$dR_t = \frac{d}{dt}\left(\int_0^t e^{-\alpha s} dW_s\right) = e^{-\alpha t} dW_t$$

Then the partial derivatives:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(y + zr) = 1$$

$$\frac{\partial f}{\partial z} = \frac{\partial}{\partial z}(y + zr) = r$$

$$\frac{\partial f}{\partial r} = \frac{\partial}{\partial r}(y + zr) = z$$

$$\frac{\partial^2 f}{\partial^2 x_i} = 0, as\ all\ first\ derivatives\ are\ constant\ respect\ to\ the\ variable$$

Then compute the quadratic variations:

$$(dY_t)^2 = 0\ (it\ is\ deterministic)$$

$$(dZ_t)^2 = 0\ (it\ is\ deterministic)$$

$$(dR_t)^2 = e^{-2\alpha t} dt$$

Lastly we replace the values:

$$dX_t = df = dY_t + R_t dZ_t + Z_t dR_t + \frac{1}{2} * 0 * e^{-2\alpha t} dt = \alpha Y_t d_t + R_t \alpha Z_t d_t + Z_t e^{-\alpha t} dW_t$$
$$= \alpha Y_t d_t + R_t \alpha Z_t d_t + e^{\alpha t}\sigma\, e^{-\alpha t} dW_t = \alpha(Y_t + R_t Z_t)\, d_t + \sigma\, dW_t$$
$$= \alpha X_t\, d_t + \sigma\, dW_t$$

2 Exercise 5.5

Suppose that X satisfies the SDE

$$dX_t = \alpha X_t dt + \sigma X_t dW_t.$$

Now define Y by $Y_t = X_t^\beta$, where β is a real number. Then Y is also a GBM process. Compute dY and find out which SDE Y satisfies.

Solution:

To solve this first we use Itô's formula with $f(x) = x^\beta$:

$$f'(x) = \beta x^{\beta-1}$$

$$f''(x) = \beta(\beta-1)x^{\beta-2}$$

And using Itô's formula the value of dYt:

$$dY_t = f'(X_t)dX_t + \frac{1}{2}f''(x)(dX_t)^2$$

From the given SDE:

$$(dX_t)^2 = (\sigma X_t dW_t)^2 = \sigma^2 {X_t}^2 d_t$$

Therefore:

$$\begin{aligned} dY_t &= \beta {X_t}^{\beta-1} dX_t + \frac{1}{2}\beta(\beta-1){X_t}^{\beta-2}\sigma^2 {X_t}^2 d_t \\ &= \beta {X_t}^{\beta-1}(\alpha X_t d_t + \sigma X_t dW_t) + \frac{1}{2}\beta(\beta-1){X_t}^{\beta}\sigma^2 d_t \\ &= \alpha\beta {X_t}^{\beta} d_t + \sigma\beta {X_t}^{\beta} dW_t + \frac{1}{2}\beta(\beta-1){X_t}^{\beta}\sigma^2 d_t \\ &= \left[\alpha\beta + \frac{1}{2}\beta(\beta-1)\sigma^2\right]{X_t}^{\beta} d_t + \beta\sigma {X_t}^{\beta} dW_t \end{aligned}$$

Since $Y_t = X_t^\beta$:

$$dY_t = \left[\alpha\beta + \frac{1}{2}\beta(\beta-1)\sigma^2\right] Y_t d_t + \beta\sigma Y_t dW_t$$

And this is a GBM with:

- Drift coefficients: $\bar{\alpha} = \alpha\beta + \frac{1}{2}\beta(\beta-1)\sigma^2$
- Diffusion coefficients: $\bar{\sigma} = \beta\sigma$

3 Exercise 5.9

Use a stochastic representation result in order to solve the following boundary value problem in the domain $[0,T] \times R$.

$$\frac{\partial F}{\partial t} + \mu(t,x)\frac{\partial F}{\partial x} + \frac{1}{2}\sigma^2(t,x)\frac{\partial^2 F}{\partial x^2} + k(t,x) = 0,$$
$$F(T,x) = \Phi(x).$$

Here μ, σ, k and Φ are assumed to be known functions. Prove that this problem has the stochastic representation formula

$$F(t,x) = E_{t,x}[\Phi(X_T)] + \int_t^T E_{t,x}[k(s,X_s)]ds,$$

where as usual X has the dynamics

$$dX_s = \mu(s,X_s)ds + \sigma(s,X_s)dW_s,$$
$$X_t = x.$$

Hint: Define X as above, assume that F actually solves the PDE and consider the process $Z_s = F(s,X_s)$.

Solution:

To solve this first we follow the hint, assume F solves the PDE and define:

$$Z_s = F(s,X_s) + \int_t^s k(\mu, X_\mu)d_\mu$$

Then we apply Itô's formula to F(s, Xs):

$$dF(s,X_s) = \frac{\partial F}{\partial t}d_s + \frac{\partial F}{\partial x}dX_s + \frac{1}{2}\frac{\partial^2 F}{\partial x^2}(dX_s)^2 = \frac{\partial F}{\partial t}d_s + \frac{\partial F}{\partial x}[\mu d_s + \sigma dW_s] + \frac{1}{2}\frac{\partial^2 F}{\partial x^2}\sigma^2 d_s$$
$$= \left[\frac{\partial F}{\partial t} + \mu\frac{\partial F}{\partial x} + \frac{1}{2}\frac{\partial^2 F}{\partial x^2}\sigma^2 d_s\right]d_s + \frac{\partial F}{\partial x}\sigma dW_s$$

Since F satisfies the SDE:

$$\left[\frac{\partial F}{\partial t} + \mu\frac{\partial F}{\partial x} + \frac{1}{2}\frac{\partial^2 F}{\partial x^2}\sigma^2 d_s\right] d_s = -k(s, X_s)$$

Therefore:

$$dF(s, X_s) = -k(s, X_s)d_s \; + \frac{\partial F}{\partial x}\sigma dW_s$$

Then we compute the dZs:

$$dZ_s = dF(s, X_s) + k(s, X_s)d_s = \frac{\partial F}{\partial x}\sigma dW_s$$

This shows Zs is a martingale (no drift term)

Integrate from t to T:

$$Z_T - Z_t = \int_t^T \frac{\partial F}{\partial x}(s, X_s)\sigma dW_s$$

Taking conditional expectation $\mathbb{E}_{t,x}[\cdot]$:

$$\mathbb{E}_{t,x}[Z_T] - Z_t = 0$$

(The stochastic integral has zero expectation)

Therefore:

$$Z_t = \mathbb{E}_{t,x}[Z_T]$$

$$F(t,x) + \int_t^t k(\mu, X_\mu)d\mu = \mathbb{E}_{t,x}\left[F(T, X_T) + \int_t^T k(\mu, X_\mu)d\mu\right]$$

$$F(t,x) = \mathbb{E}_{t,x}[F(T, X_T)] + \mathbb{E}_{t,x}\left[\int_t^T k(\mu, X_\mu)d\mu\right]$$

Using $F(T, x) = \Phi(x)$and Fubini's theorem:

$$F(t,x) = \mathbb{E}_{t,x}[\Phi(X_T)] + \int_t^T \mathbb{E}_{t,x}[k(s,X_s)]d_s$$

4 Exercise 7.3 (Modified)

Consider the standard Black–Scholes model. Derive the Black–Scholes formula for the European **put** option.

Solution:

Setup:

- Stock: $dS_t = \mu S_t dt + \sigma S_t dW_t, S_0 = s$
- Risk-free rate: r
- Put payoff: $(K - S_T)^+$

Find: Put price $\mathrm{P(t,s)} = e^{-r(T-t)}\mathbb{E}_t{}^Q[(K - S_T)^+]$

Under the risk-neutral measure Q, the stock follows:

$$dS_t = rS_t d_t + \sigma S_t d\widetilde{W_t}$$

The solution is:

$$S_T = S_t e^{\left(r-\frac{1}{2}\sigma^2\right)(T-t)+\sqrt{T-t}Z}, where\ Z{\sim}N(0,1)$$

Put–Call Parity:

$$\mathrm{P}(t,s) + S_t = \mathrm{C}(t,s) + Ke^{-r(T-t)}, where\ \mathrm{C}(t,s)\ is\ the\ call\ price$$

Black-Scholes Call formula:

$$\mathrm{C}(t,s) = S_T\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2)$$

$$d_1 = \frac{ln\left(S_t/K\right) + \left(r + 1/2\,\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$$

$$d_2 = d_1 - \sigma\sqrt{T-t} = \frac{ln\left(S_t/K\right) + \left(r - 1/2\,\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$$

Using Put-Call Parity:

$$\mathrm{P}(t,s) = \mathrm{C}(t,s) + Ke^{-r(T-t)} - S_t = S_T\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2) + Ke^{-r(T-t)} - S_t$$
$$= S_t[\Phi(d_1) - 1] + Ke^{-r(T-t)}[1 - \Phi(d_2)]$$

Using $\Phi(-x) = 1 - \Phi(x)$:

$$\mathrm{P}(t,s) = Ke^{-r(T-t)}\Phi(-d_2) - S_t[-\Phi(d_1)]$$

5 Exercise 7.4

Consider the standard Black–Scholes model. Derive the arbitrage free price process for the T-claim $\mathcal{X}$ where $\mathcal{X}$ is given by $\mathcal{X} = S(T)^{\beta}$. Here β is a known constant.
Hint: For this problem you may find Exercises 5.5 and 4.4 useful.

Solution:

To solve this first under the risk-neutral measure Q,:

$$dS_t = rS_t d_t + \sigma S_t d\widetilde{W}_t$$

Using Itô's formula, if $X_t = S_t^{\beta}$

$$dX_t = \left[\beta r + \frac{1}{2}\beta(\beta-1)\sigma^2\right]X_t d_t + \beta\sigma X_t d\widetilde{W}_t$$

Then we solve for Xt, since it follows a Geometric Brownian Motion:

$$X_T = X_t e^{\left\{\left[\beta r + \frac{1}{2}\beta(\beta-1)\sigma^2 - \frac{1}{2}\beta^2\sigma^2\right](T-t) + \beta\sigma\sqrt{T-t}Z\right\}}, where\ Z \sim N(0,1)$$

Then:

$$X_T = {S_t}^{\beta} e^{\left\{\left[\beta r + -\frac{1}{2}\beta\sigma^2\right](T-t) + \beta\sigma\sqrt{T-t}Z\right\}}, where\ Z \sim N(0,1)$$

Lastly we price at time t using the arbitrage-free price:

$$V(t,S_t) = e^{-r(T-t)}\mathbb{E}_t{}^{Q}\left[{S_T}^{\beta}\right] = e^{-r(T-t)}{S_t}^{\beta}\mathbb{E}_t{}^{Q}\left[e^{\left\{\left[\beta r + -\frac{1}{2}\beta\sigma^2\right](T-t) + \beta\sigma\sqrt{T-t}Z\right\}}\right]$$

Using: $\mathbb{E}[e^{aZ}] = e^{\frac{a^2}{2}}$

Then:

$$V(t,S_t) = {S_t}^{\beta} e^{\left\{-r(T-t) + \beta r(T-t) - \frac{1}{2}\beta\sigma^2(T-t) + \frac{1}{2}\beta^2\sigma^2(T-t)\right\}} = {S_t}^{\beta} e^{\left\{r(\beta-1)(T-t) + \frac{1}{2}\beta(\beta-1)\sigma^2(T-t)\right\}}$$
$$= {S_t}^{\beta} e^{\left\{\left[r(\beta-1) + \frac{1}{2}\beta(\beta-1)\sigma^2\right](T-t)\right\}}$$